F79SP Session 17: Branching processes: Extinction probability

1 The probability of extinction

We first discuss the following cases:

1. $p_0 = 0$, $p_1 < 1$.
   - $X_{n+1} \geq X_n$ for all n.
   - $i \to j$ if and only if $i \leq j$, so each state $i \geq 1$ is transient.
   - State 0 is absorbing but it is not reachable from any other states.
   - In this case, extinction cannot happen if the initial population is non-zero.
2. $p_0 > 0$, $p_0 + p_1 < 1$.
   - Note that population can disappear in one step, with probability $p_{i0} = p_0^i > 0$.
   - Hence, 0 is absorbing, and all states $i \geq 1$ are transient, that is, such State i will only be visited finitely many times.
   - This means, the branching process must either be absorbed into 0, or $X_n \to \infty$ as $n \to \infty$.
   - We are interested with the probability of being absorbed into 0 (The extinction probability).

Key idea: The study the extinction probability starting from $X_0 = i$ can be reduced to the study of extinction probability of a branch, $X_0 = 1$.

- Let D be the event of "death", or extinction:

$$D := \{X_n = 0 \text{ for some } n \geq 0\}.$$

- We wish to compute the extinction probability

$$\varepsilon(i) = \mathbf{P}_i(D) = \mathbf{P}(D|X_0 = i).$$

- We show that:

$$\varepsilon(i) = \varepsilon^i \text{ where } \varepsilon = \varepsilon(1) = \mathbf{P}_1(D).$$

Indeed, if we start with $X_0 = i \geq 1$ initial ancestors, each one behaves independently of one another. For each $l = 1, ..., i$, let D_l be the event that the branching process corresponding to the lth initial ancestor dies. We then have that

$$D = D_1 \cap D_2 \cap \cdots \cap D_i$$

so

$$\mathbf{P}_i(D) = \mathbf{P}(D|X_0 = i) = \mathbf{P}(D_1) \cdot \cdots \cdot \mathbf{P}(D_i) = \varepsilon^i.$$

The following theorem illustrates on how to determine $\varepsilon = \varepsilon(1)$.

Theorem 1.1. *Consider a Galton-Watson process starting from $X_0 = 1$ initial ancestor. Let*

$$\mu = \mathbf{E}\,\xi = \sum_{k=1}^{\infty} k p_k$$

be the mean number of offspring of an individual. Assume that $p_0 > 0$ and $p_0 + p_1 < 1$. Let

$$\varphi(z) = \mathbf{E}\, z^{\xi} = \sum_{k=0}^{\infty} z^k p_k$$

be the probability generating function of ξ. Let ε be the extinction probability of the process. Then

1. *if* $\mu \leq 1$ *then the* $\varepsilon = 1$ *and*
2. *if* $\mu > 1$ *then the 'fixed-point equation'*

$$\varphi(z) = z,$$

has only two solutions, $0 < z_1 < z_2 = 1$*, and* $\varepsilon = z_1$ *is the minimal one.*

We go through the following examples.

Example 1.1. *Consider a Galton-Watson process as follows:*

1. *Suppose that* $X_0 = 3$*,* $p_0 = \frac{2}{3}$*,* $p_1 = \frac{1}{4}$*, and* $p_3 = \frac{1}{12}$*. Find extinction probability.*
2. *Suppose that* $X_0 = 4$*,* $p_0 = \frac{1}{3}$*,* $p_1 = \frac{1}{6}$*, and* $p_2 = \frac{1}{2}$*. Find extinction probability.*

Justify extinction probability via Markov chain perspective

- First, recall that $\epsilon(i) = \epsilon^i$.
- Note that $\epsilon = \epsilon(1)$ is the absorption probability into the closed class $\{0\}$, given starting at $X_0 = 1$.
- Using conditioning on first step method:

- We can further show that the extinction probability is the minimum solution to the fixed point equation above (see notes).

Justify extinction probability via random walk perspective

- Lets modify dynamics of the branching process by assuming that individuals bear offsprings one-by-one, within each generation.
- This gives a random walk with increments having the distribution of $\xi - 1$.
- Here, extinction probability can be interpreted as the ruin probability $P(R)$, which we know $P(R) < 1$ if and only if $\mathbf{E}\xi - 1 > 0$.